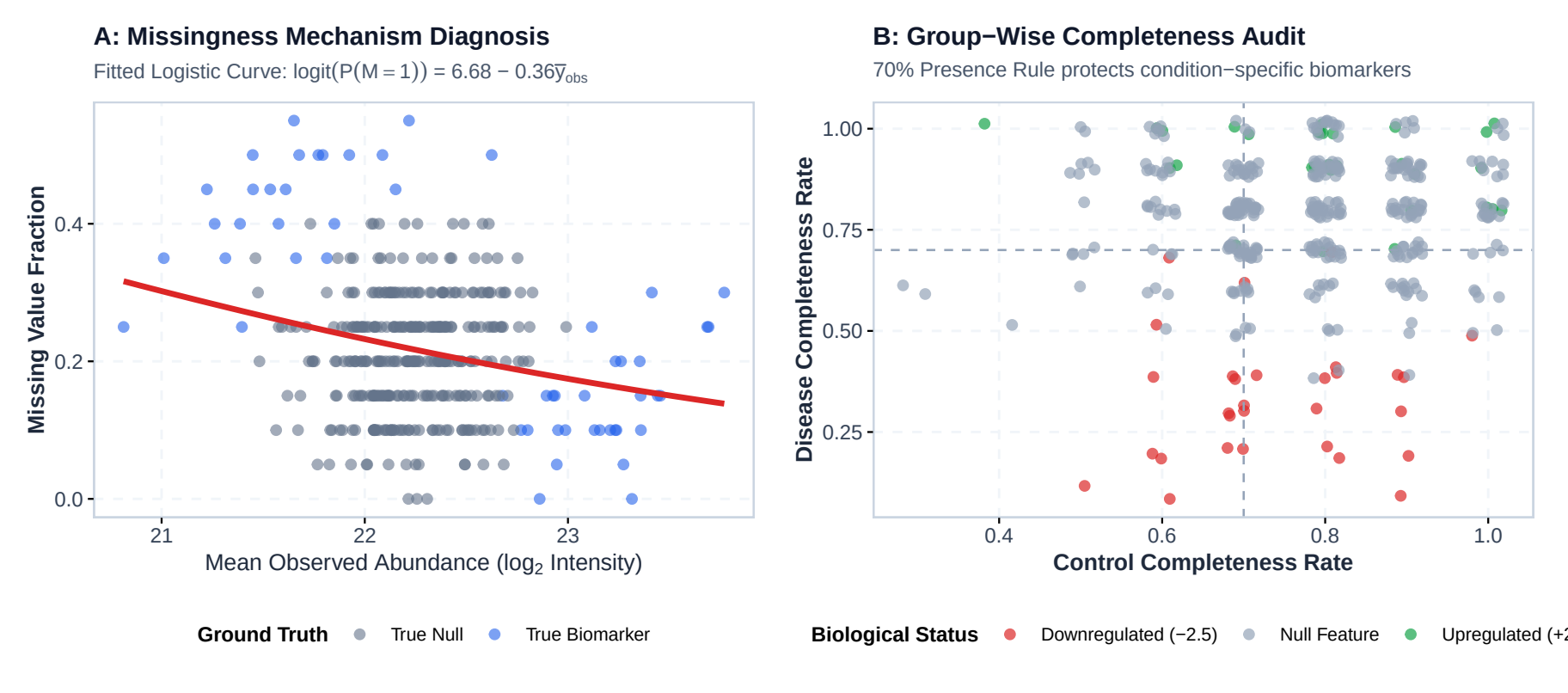
A: Missingness Mechanism Diagnosis
Fitted Logistic Curve: logit(P(M = 1)) = 6.68 − 0.36$\bar{y}_{obs}$
Missing Value Fraction
0.0
0.2
0.4
21
22
23
Mean Observed Abundance (log$_2$ Intensity)
Ground Truth
True Null
True Biomarker
B: Group−Wise Completeness Audit
70% Presence Rule protects condition−specific biomarkers
Disease Completeness Rate
0.25
0.50
0.75
1.00
0.4
0.6
0.8
1.0
Control Completeness Rate
Biological Status
Downregulated (−2.5)
Null Feature
Upregulated (+2